

Model Enumeration using IPASIR-UP without Blocking Clauses

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Model Enumeration

cnf: $(X \vee Y) \wedge (\bar{Y} \vee Z)$

solution 1 : $X \wedge Y \wedge Z$

solution 2 : $\bar{X} \wedge Y \wedge Z$

solution 3 : $X \wedge \bar{Y} \wedge Z$

solution 4 : $X \wedge \bar{Y} \wedge \bar{Z}$

Model Enumeration using Blocking Clauses

Pseudo Code

```
list[model] ms
while solver.solve(cnf) == SAT:
    model m = solver.solution()
    ms += [m]
    cnf += not(m)
```

Model Enumeration using Blocking Clauses

Blocking Clause

$$cnf : (X \vee Y) \wedge (\bar{Y} \vee Z)$$

$$model : X \wedge Y \wedge Z$$

$$\text{not}(model) : \neg(X \wedge Y \wedge Z) \iff \bar{X} \vee \bar{Y} \vee \bar{Z}$$

$$new\ cnf : (X \vee Y) \wedge (\bar{Y} \vee Z) \wedge (\bar{X} \vee \bar{Y} \vee \bar{Z})$$

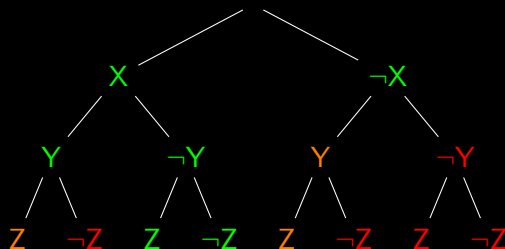
Model Enumeration using IPASIR-UP

IPASIR-UP extends IPASIR by an User Propagator

- ▶ `cb_check_found_model`
- ▶ `cb_has_external_clause`
- ▶ `cb_add_external_clause_lit`
- ▶ `notify_assignment`
- ▶ `notify_new_decision_level`
- ▶ `notify_backtrack`
- ▶ `cb_decide`
- ▶ `cb_propagate`
- ▶ `cb_add_reason_clause_lit`

Model Enumeration without Blocking Clauses

cnf: $(X \vee Y) \wedge (\bar{Y} \vee Z)$



Model Enumeration without Blocking Clauses

Continue the search after finding a model

```
bool cb_check_found_model(model &m):  
    ...  
    new_dl = highest_dl_with_positive_decision()  
    if (new_dl <= 0) return true  
    solver->force_backtrack(new_dl - 1)  
    ...  
    return false
```

Model Enumeration without Blocking Clauses

Continue the search after backtracking

1. Propagate the negation of the old decision
2. Decide the negation of the old decision

Model Enumeration without Blocking Clauses

Propagate the negation of the old decision

```
bool cb_propagate():  
    ...  
    return -decisions.back()  
  
bool cb_add_reason_clause_lit(propagated_lit):  
    return propagated_lit  
    return -decisions[dl]  
    ...  
    return -decisions[1]  
    return 0
```

Model Enumeration without Blocking Clauses

Decide the negation of the old decision

```
bool cb_decide():  
    ...  
    if decision_count_per_level[dl + 1] == 1:  
        return -decisions[dl + 1]  
    ...
```

In `cb_check_found_model`:

```
# new_dl = highest_dl_with_positive_decision()  
new_dl = highest_dl_with_decision_count_one()
```

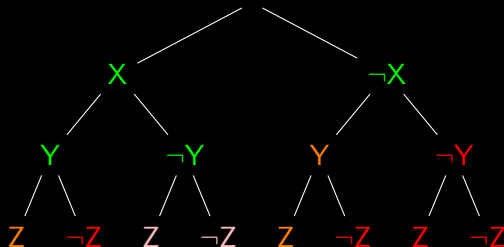
Model Enumeration without Blocking Clauses

CaDiCaL may:

1. propagate the previous decision
2. encounter a conflict during propagation
3. propagate only literals different from the previous decision
4. propagate the negation of the previous decision

Implicant Shrinking

cnf: $(X \vee Y) \wedge (\bar{Y} \vee Z)$



Implicant shrinking

2-watching literal scheme

$$\text{cnf} : \quad (\overset{\uparrow}{X} \vee \overset{\uparrow}{Y}) \wedge (\overset{\uparrow}{\bar{Y}} \vee \overset{\uparrow}{Z})$$

$$\text{cnf}' : \quad (\overset{\uparrow}{X} \vee \overset{\uparrow}{Y}) \wedge (\overset{\uparrow}{\bar{Y}} \vee \overset{\uparrow}{Z} \vee X)$$

Implicant shrinking

Access to watches

```
int check_literal(int v, int b, ...):  
    for lit in other_watching_literal(v):  
        if !sat(lit, get_n(stack, index)):  
            b = max(b, decision_level(1))  
    return b
```

Drawbacks

- ▶ Maintaining chronological order (`ask_decision`)
- ▶ Out of order propagations
- ▶ No access to watches
- ▶ Optimizations: Interacting with the solver
- ▶ Optimizations: Duplication of data

Benchmarks

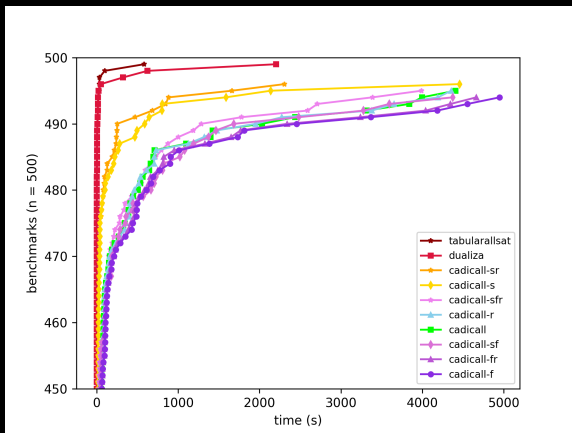
Benchmark Sets

- ▶ TabularAllSat
- ▶ SATLIB
- ▶ Model Counting Competition 2025

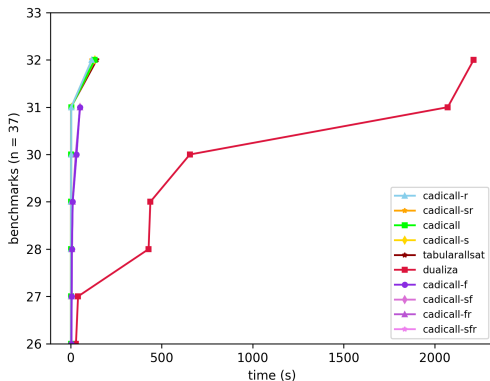
Solvers

- ▶ CaDiCAll
- ▶ TabularAllSat
- ▶ Dualiza

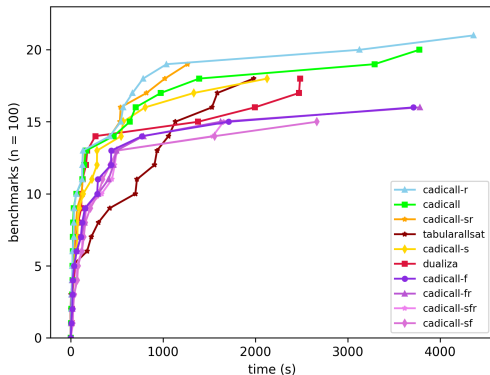
TabularAllSat Benchmarks



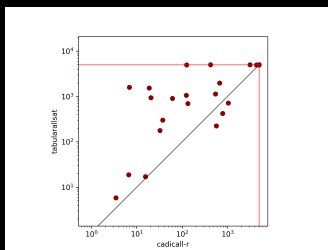
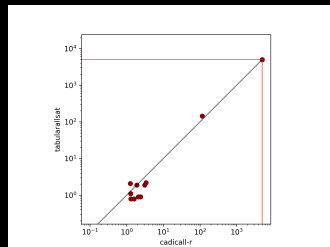
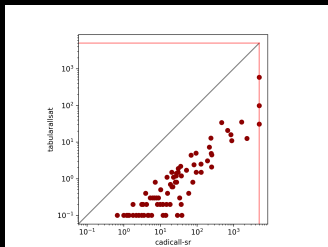
SATLIB Benchmarks



Model Counting Competition 2025 Benchmarks



Four Figures - BMS, Custom, MC



Mismatching Results

TabularAllSat Benchmarks (T C vs D)

- ▶ BMS_k3_n100_m429_454

SATLIB Benchmarks (T vs C D)

- ▶ qg4-09

Model Counting Competition 2025 Benchmarks (T vs C D)

- ▶ mc2025_track1_79
- ▶ mc2025_track1_93
- ▶ mc2025_track1_95
- ▶ mc2025_track1_101

Thats about it

Showcase?